

6G Weekly Survey — Sep 21 - Sep 27, 2026

15 items across 6 topics, ranked by importance.

6G Demos & Trials

Agentic RDZ: Autonomous Zone Management with AI Agents and an FR3 Coexistence Use Case

Minh Dat Nguyen, Gabriele Gemmi, Tamerlan Aghayev, Paolo Testolina, et al. · arXiv cs.NI · 2026-09-15

★★★★■ Addresses a critical 6G challenge (FR3 spectrum sharing and dynamic spectrum access) with a novel agentic architecture validated on real O-RAN hardware, demonstrating practical viability for flexible spectrum management in mid-band 6G deployments.

This paper introduces an agentic Radio Dynamic Zone (A-RDZ) that uses Large Language Models to autonomously manage spectrum sharing between experimental 5G NR systems and incumbent satellite users in FR3 (7-24 GHz). The system combines LLM-based reasoning for policy interpretation and experiment orchestration with deterministic safeguards and near-real-time reflexes to guarantee incumbent protection. Hardware-in-the-loop validation on an O-RAN testbed demonstrates end-to-end operation: detecting emission violations from live spectrum measurements, selecting mitigation strategies, and applying corrective actions through O-RAN control interfaces.

Contribution: First demonstration of LLM-based autonomous agents managing real-time spectrum coexistence in FR3 bands with hardware-validated detection-to-mitigation workflows on an operational O-RAN system.

Relevance to your work: Directly relevant to your beyond-5G work on AI-native air interfaces and next-generation systems, particularly for autonomous spectrum management, O-RAN integration, and FR3 bands that will be central to 6G deployments requiring coexistence with satellite incumbents.

Bilevel Optimization of Topology and Hyperparameters (BOTH)

Suryanarayanan Manoj Sanu, Miguel Anibal Bessa, Alejandro Marcos Aragón · arXiv cs.LG · 2026-09-18

★☆☆■■ While technically sound for computational mechanics and design automation, this work addresses topology optimization for structural engineering rather than wireless system design or air interface optimization.

This paper addresses the challenge of hyperparameter tuning in topology optimization (TO) by proposing a bilevel optimization approach that uses automatic differentiation to compute hypergradients, enabling simultaneous optimization of both design topology and hyperparameters. The method evaluates one or two TO steps to inform hyperparameter updates and scales to thousands of hyperparameters with cost comparable to a few standard TO runs. Demonstrations include stress-constrained problems and compliance optimization with neural density field parameterization.

Contribution: Novel use of automatic differentiation to derive hypergradients for bilevel optimization of topology and hyperparameters simultaneously, eliminating manual tuning of 'magic numbers' in topology optimization workflows.

Relevance to your work: tangential

Multi-Domain Clustering via Measure Quantization

Rafael Pereira Eufrazio, Eduardo Fernandes Montesuma, Charles Casimiro Cavalcante · arXiv cs.LG · 2026-09-18

★★■■■ Competent machine learning methodology for multi-domain data, but addresses generic clustering rather than wireless-specific problems like channel modeling, CSI processing, or PHY-layer learning.

The paper proposes a multi-domain clustering framework that learns shared cluster prototypes across multiple data domains by minimizing probability metrics (Sinkhorn divergence or Maximum Mean Discrepancy) between domain distributions and prototype distributions. Cluster assignment uses either nearest centroid or optimal transport coupling, with mini-batch optimization enabling scalability. Experiments

on image, audio, and sensor benchmarks demonstrate consistent improvements over classical and multi-domain baselines at scale.

Contribution: Novel use of measure quantization with probability metrics (particularly Sinkhorn divergence) to perform clustering across heterogeneous domains with shared prototypes, enabling collaborative assignment via optimal transport.

Relevance to your work: Tangential—while multi-domain learning could theoretically apply to heterogeneous wireless environments (e.g., clustering channel states across multiple cells or frequency bands), the paper focuses on general ML benchmarks without wireless applications or physical-layer context.

AutoRecLab: Describe the Experiment, Get the Code!

Moritz Baumgart, Philipp Meister, Justus Krell, Michael Schmidt, et al. · arXiv cs.LG · 2026-09-18

★☆☆☆☆ While automation of ML experiments is valuable, this work targets recommender systems and has no connection to wireless physical layer, channel modeling, or 6G air-interface research.

AutoRecLab is a Python-based autonomous lab that generates executable recommender-systems code from natural-language experiment descriptions, using retrieval-augmented generation, static type checking, and execution-guided tree search. The system autonomously implements experiments (e.g., explicit-to-implicit feedback conversion) across multiple algorithms and datasets. Demonstration shows 8/9 successful runs at ~\$1 per run using GPT-4-mini.

Contribution: First autonomous code-generation framework for recommender-systems research that translates natural-language prompts into complete, executable experiments with iterative validation.

Relevance to your work: tangential

Internal-Model-Control-Based Virtual Admittance Emulation for Enhanced Grid-Forming Performance

Ruzica Cvetanovic, Lazar Stojanovic, Paolo Sbabo, Paolo Mattavelli, et al. · arXiv eess.SY · 2026-09-18

★☆☆☆☆ This work focuses on power grid converter control for renewable energy systems, which is entirely outside the scope of wireless communications and 6G research.

This paper addresses grid-forming converters for renewable energy integration by proposing an internal-model-control-based virtual admittance control that accounts for inner current-loop effects and voltage feedforward. The method ensures converter terminal admittance matches target admittance outside outer-loop frequencies, preventing small-signal harmonic instabilities that arise in state-of-the-art approaches when using high-cutoff voltage feedforward filters. Experimental validation on three-phase voltage-source converters demonstrates stable operation across feedforward filter configurations.

Contribution: A novel virtual admittance control design that inherently compensates for inner-loop dynamics, eliminating harmonic-range instabilities present in conventional approaches.

Relevance to your work: tangential

Purification and Regulation: Comorbidity-Aware Multi-Label Few-Shot Learning for Medical Image Classification

Ying-Chih Lin, Po-Chih Kuo, Yong-Sheng Chen · arXiv cs.LG · 2026-09-18

★☆☆☆☆ While technically sound for medical imaging, this work focuses on computer vision meta-learning for clinical diagnosis with no connection to wireless communication, physical layer design, or 6G system architecture.

This paper addresses multi-label few-shot learning for medical image classification by proposing Prototype Purification and Regulation (PPR), which generates cleaner disease prototypes by filtering comorbidity information and adaptively regulates inter-class distances based on disease correlation statistics. The method is validated on four chest X-ray datasets showing improved disease detection and cross-domain generalization. PPR uses sample-level and disease-level comorbidity scores to create a comorbidity-aware embedding space that better captures disease-specific features and their relationships.

Contribution: The novel contribution is the dual-stage approach of purifying prototypes from entangled comorbidity information at the sample level, then adaptively regulating inter-class prototype distances using disease-level comorbidity statistics, rather than enforcing uniform separability.

Relevance to your work: tangential

6G Physical Layer

Ambiguity Function Analysis of OFDM Signals With Pilots and Data Payloads

Jialin Wu, Fan Liu, Ying Zhang, Yifeng Xiong, et al. · arXiv eess.SP · 2026-09-15

★★★★■ Directly addresses a fundamental ISAC design conflict with mathematical rigor, showing that 5G/6G pilot structures optimized for channel estimation degrade sensing performance—a critical insight for dual-function waveform design.

This paper analyzes the ambiguity function (AF) of practical OFDM frames containing both deterministic pilots and random data when reused for integrated sensing and communication (ISAC). It derives closed-form expressions for expected sidelobe levels under two Doppler regimes (DP-AF and FST-AF), proving that communication-optimized pilot patterns like Zadoff-Chu sequences produce pronounced delay-Doppler ridges and periodic sidelobe structures unsuitable for radar estimation. The authors demonstrate that irregular pilot placement can suppress high sidelobes and improve target estimation performance.

Contribution: First rigorous characterization showing that standard communication pilot patterns (contiguous or equally-spaced ZC sequences) inherently create high ambiguity sidelobes in ISAC scenarios, with proof that no pilot design minimizes all sidelobes simultaneously.

Relevance to your work: Directly relevant to your ISAC research: provides theoretical foundation for redesigning OFDM pilot patterns to balance communication channel estimation with radar ambiguity function requirements in AI-native air interfaces.

The Price of the Golden 6G Band: Evaluation of Beam Management Effort in FR3

Clémence Altmeyerhenzien, Ljiljana Simić, Marina Petrova · arXiv cs.NI · 2026-09-15

★★★★■ Addresses critical practical deployment question for 6G FR3 spectrum with empirical analysis that will inform 3GPP beam management design and infrastructure planning decisions.

This paper evaluates beam management overhead in FR3 (7.125-24.25 GHz), the proposed 6G "golden band," through urban network simulations. The authors quantify beam alignment sensitivity, handover rates, beam switching frequency, and steering distances across FR3 to assess feasibility of reusing FR1 infrastructure with beamforming. Results show that while lower FR3 frequencies benefit from wider beams reducing tracking stringency, non-adjacent beam switching rates remain comparable to FR2 (mmWave), indicating substantial beam management overhead persists.

Contribution: First comprehensive characterization of beam management complexity specifically for FR3 frequencies, quantifying the practical overhead of deploying directional communication in the mid-band spectrum proposed for 6G.

Relevance to your work: Directly relevant to your beam management research interests; provides baseline complexity metrics for FR3 that contextualize your AI/ML beam prediction and tracking work for next-generation systems.

Secure RIS-Aided Multicasting: Globally Optimal Beam Management and Discrete-Phase RIS Configuration

Luis F. Abanto-Leon, Setareh Maghsudi · arXiv cs.IT · 2026-09-18

★★★★■ Solid incremental work on RIS physical-layer security with rigorous optimization, but limited to narrow multicast security scenario without broader AI/ML or next-generation waveform integration.

This paper addresses security in RIS-aided wireless multicasting by jointly optimizing discrete-phase RIS configurations and base station beam management to minimize eavesdropper signal quality. The authors reformulate a nonconvex mixed-integer nonlinear program into a tractable mixed-integer quadratically constrained program that achieves global optimality. Numerical results show up to 58% reduction in wiretap SNR compared to baselines.

Contribution: First globally optimal solution for joint RIS phase-shift and BS beam management that accounts for discrete phase constraints and colluding eavesdroppers in secure multicasting.

Relevance to your work: Tangential—while RIS beam management relates to your massive MIMO and beam management interests, the focus on physical-layer security and discrete optimization does not directly connect to AI-native air interfaces or learned PHY approaches.

ISAC (sensing + communication)

SemiISAC: Semantic Integrated Sensing and Communications

Xiaoqi Zhang, J. Andrew Zhang, Zhongqin Wang, Chang Liu, et al. · arXiv eess.SP · 2026-09-16

★★★★■ Addresses a fundamental limitation in current ISAC research by introducing semantic-level integration with rigorous theoretical foundations and practical neural network realizations, though experimental validation remains limited to simulation.

This paper proposes semantic ISAC (SemiISAC), a framework that extends integrated sensing and communication beyond physical-layer resource sharing by explicitly incorporating task-relevant semantic information from both communication and sensing tasks. The authors provide an information-theoretic characterization of the semantic trade-off and demonstrate two optimization approaches: end-to-end learning and modular optimization. A concrete OFDM realization with learnable time-frequency precoders shows improved communication-sensing trade-offs compared to physical-layer baselines.

Contribution: First framework to unify semantic communication and semantic sensing in ISAC, moving beyond physical-layer integration to explicitly exploit task-relevant information with information-theoretic grounding.

Relevance to your work: Directly combines three of your core research areas—ISAC, semantic/task-oriented communication, and AI/ML for PHY—providing a principled framework for joint optimization that could inform your work on AI-native air interfaces for 6G.

SARAMS: Split-Aware Joint Resource Allocation for Multi-Monostatic Sensing

Maximiliano Rivera Figueroa, Pradyumna Kumar Bishoyi, Marina Petrova · arXiv eess.SP · 2026-09-18

★★★★■ Addresses a critical but overlooked practical constraint (3GPP split architecture) in multi-BS cooperative sensing, bridging the gap between theoretical ISAC localization and real network deployments with rigorous optimization framework.

The paper addresses multi-monostatic ISAC localization where base stations fuse observations at a central unit, explicitly modeling how 3GPP functional splits constrain fronthaul bitrate and determine whether coherent or non-coherent fusion is feasible. SARAMS jointly optimizes per-BS power, bandwidth, and observation time to minimize worst-case multi-target position error bounds under fronthaul, computational, and power constraints, formulated as a mixed-integer nonlinear program solved via semidefinite relaxation and block coordinate descent. Simulations show 65.7% reduction in 90th-percentile worst-case position error bound versus equal power allocation with 8.7% optimality gap.

Contribution: First work to explicitly model 3GPP functional split constraints (fronthaul bitrate, computational load) and their impact on coherent versus non-coherent fusion when optimizing multi-monostatic ISAC localization accuracy.

Relevance to your work: Directly relevant to your ISAC research: demonstrates how fronthaul-limited distributed architectures constrain sensing fusion strategies, a practical consideration for any multi-BS JCAS deployment in beyond-5G systems.

NTN / satellite

AI-based Interference Mitigation for Power-domain Spectrum Sharing among LEO Satellites

Bowen Zhang, Barry Evans, Pei Xiao · arXiv eess.SP · 2026-09-14

★★★★■ Addresses a critical practical problem (LEO co-channel interference) with growing commercial urgency as mega-constellations proliferate, validated with real Starlink/OneWeb parameters.

The paper addresses co-channel interference during In-Line-Events (ILE) between LEO constellations (e.g., Starlink and Eutelsat OneWeb) operating in Ku-band. It proposes coordinated transmit power control to create sufficient received-power gaps, enabling user-side successive interference cancellation (SIC) or AI-assisted receivers that require smaller power differentials. Evaluation with real constellation waveforms demonstrates improved interference mitigation and spectrum efficiency.

Contribution: Novel power-domain coordination scheme for multi-operator LEO spectrum sharing that enables practical SIC or AI receivers by creating intentional power gaps during interference events.

Relevance to your work: Directly relevant: combines NTN systems with AI-assisted physical-layer interference mitigation, providing a concrete use case for neural receivers in commercial LEO deployments where traditional SIC struggles.

AI-native air interface

Pinching Antennas for Next-Generation Wireless Communications: A Network Perspective

Yanqing Xu, Shan Shan, Yongxu Zhu, Zhiguo Ding, et al. · arXiv eess.SP · 2026-09-15

★★★★■ Intriguing concept with network-level implications, but physical realizability and practical constraints remain unclear; tutorial nature suggests early-stage exploration rather than validated breakthrough.

The paper introduces pinching antennas, which dynamically reconfigure radiation point locations along waveguides after deployment, creating a spatial degree of freedom beyond conventional beamforming. It develops network-level design frameworks including environment-division multiple access, multi-cell coordination, and cell-free architectures that exploit radiation-location reconfigurability for interference management and traffic adaptation. System models, optimization formulations, and deployment modes (standalone, BS-integrated, distributed) are presented with future directions in mobility, AI-native control, NTN integration, and ISAC.

Contribution: Novel concept of radiation-location reconfigurability as a network resource, moving beyond fixed antenna positions to enable post-deployment spatial optimization at the network level rather than just beamforming at the link level.

Relevance to your work: Connects to AI-native air interface through proposed AI-native control frameworks and to ISAC through identified sensing integration opportunities, though physical layer implementation details for these applications are not yet developed.

AI-Native Open RAN: A Roadmap from xApps and rApps to Autonomous Network Agents

Ryan Barker, Alireza Ebrahimi Dorcheh, Tolunay Seyfi, Mohammad Raihan Uddin, et al. · arXiv cs.NI · 2026-09-14

★★★★■ Solid survey work that organizes existing approaches and identifies the generalization gap, but remains largely a literature review without novel methods, datasets, or deployment insights that would directly advance AI-native physical layer design.

This paper surveys AI integration in Open RAN architectures, reviewing how machine learning, deep reinforcement learning, digital twins, and foundation models address RAN optimization tasks like resource management, slicing, and interference mitigation. The authors provide a taxonomy of AI approaches across near-real-time and non-real-time RIC frameworks, highlighting current limitations in generalization and cross-environment deployment. They propose a roadmap toward autonomous network agents that move

beyond task-specific xApps/rApps to more generalizable AI-native solutions.

Contribution: The work synthesizes the fragmented O-RAN AI literature into a unified taxonomy and critiques the lack of generalization in existing solutions, positioning foundation models and autonomous agents as the path toward truly AI-native RAN intelligence.

Relevance to your work: Tangential—focuses on upper-layer RAN optimization (resource management, slicing) rather than the physical-layer AI/ML (neural receivers, learned PHY, ISAC) and channel modeling central to your AI-native air interface research.

Semantic communication

Goal-Oriented Communication and Control Co-Design via Semantic Push-Pull in Industrial IoT

Muhammad Azeem Khan, Yuriy Zacchia Lun, Aamir Mahmood, Piergiuseppe Di Marco, et al. · arXiv cs.IT · 2026-09-18

★★★★ Solid application of semantic communication principles to industrial control with practical co-design insights, though limited to simulation validation and incremental over existing event-triggered control literature.

This paper tackles wireless networked control in industrial IoT by introducing a semantic-layer framework that replaces age-of-information metrics with a state-to-error ratio (SER) capturing actual control impact. The system combines threshold-based sensor push and state-aware controller pull mechanisms, scaling actuation deadbands by plant dynamics to ensure fair resource allocation. Simulations show tracking accuracy matching periodic schedulers at lower overhead while avoiding pure event-triggered estimation failures.

Contribution: Novel semantic metric (SER) unifying control confidence and SNR to enable goal-oriented uplink/downlink arbitration that shifts the Pareto frontier between communication rate and control quality in industrial wireless control systems.

Relevance to your work: Directly relevant as a concrete instantiation of task-oriented communication for industrial verticals, demonstrating how semantic abstractions (SER vs. AoI) enable resource-efficient 6G IoT use cases with measurable performance gains.